Radio over Fiber (RoF)

Transport of radio waveforms over optical fiber: signal path, architectures, alternatives and limits

Technical review · 15 minutes · two speakers

Speaker A: problem, definition, signal path, architectures, RoF types

Speaker B: comparisons, benefits and conditions, limitations, case, conclusions



Problem and scope

Engineering problem

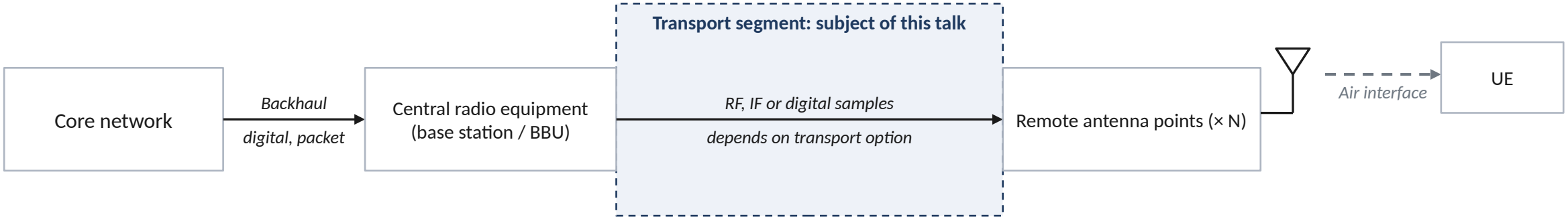
- Tunnels, stadiums and large buildings cannot be covered from one antenna site; radio power must be distributed over many antenna points.
- Radio equipment is often centralized (equipment room or base-station hotel) to share power, cooling, maintenance and multi-operator access.
- Question: how to carry the radio signal between the central equipment and each antenna point, over tens of metres to several kilometres, within a loss, noise, distortion and latency budget.

Scope of this talk

In scope: the transport segment between central equipment and remote antenna units (fronthaul or DAS feeder), downlink and uplink.

Out of scope: backhaul to the core network, air-interface design, spectrum and radio planning.

Alternatives compared: passive coaxial feeder; digital fronthaul (CPRI / eCPRI).



Downlink direction shown; uplink is the reverse path.

Definition: what RoF transports, and what it does not

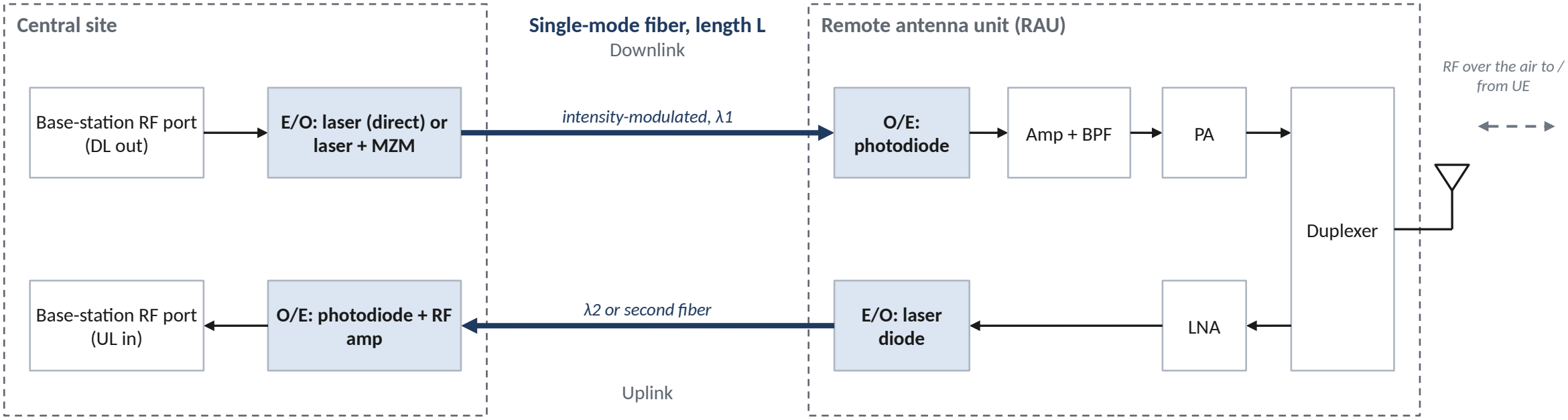
Radio over Fiber (RoF): an optical link that carries an analog radio waveform, at RF or IF, using electrical-to-optical (E/O) conversion at one end and optical-to-electrical (O/E) conversion at the other. The radio signal is generated and demodulated outside the link.

Term	Meaning used in this talk
RF	Waveform at the air-interface carrier frequency; radiated after amplification and filtering.
IF	Same modulation on a lower carrier; the remote unit needs a mixer and local oscillator (LO) to reach RF.
Baseband (I/Q)	Complex envelope centred at 0 Hz; reaching RF needs a DAC and up-conversion. Digital fronthaul carries it as samples.
Fronthaul / backhaul	Fronthaul: central baseband or radio site ↔ remote radio or antenna unit. Backhaul: base station ↔ core network.
Transport vs wireless capacity	Transport: what the link carries (bit rate, or analog bandwidth and dynamic range). Wireless: user throughput, set by spectrum, SINR, MIMO layers and cell density.
Analog RoF vs digital fronthaul	Analog waveform on fiber, versus serialized digital samples (CPRI) or packets (eCPRI). Some literature calls the latter "digital RoF".

RoF moves the point where the radio signal is radiated. It does not add spectrum or air-interface capacity.

Sources: ITU-T G.Suppl.55 (10/2025), Radio-over-fibre technologies and their applications; ITU-T G.9803 (2018), Radio over fibre systems; CPRI Specification V7.0 (2015); eCPRI Specification V2.0 (2019).

End-to-end signal path: downlink and uplink



Conversion points
DL: E/O at central site, O/E at RAU. UL: E/O at RAU, O/E at central site. Each conversion adds noise and distortion.

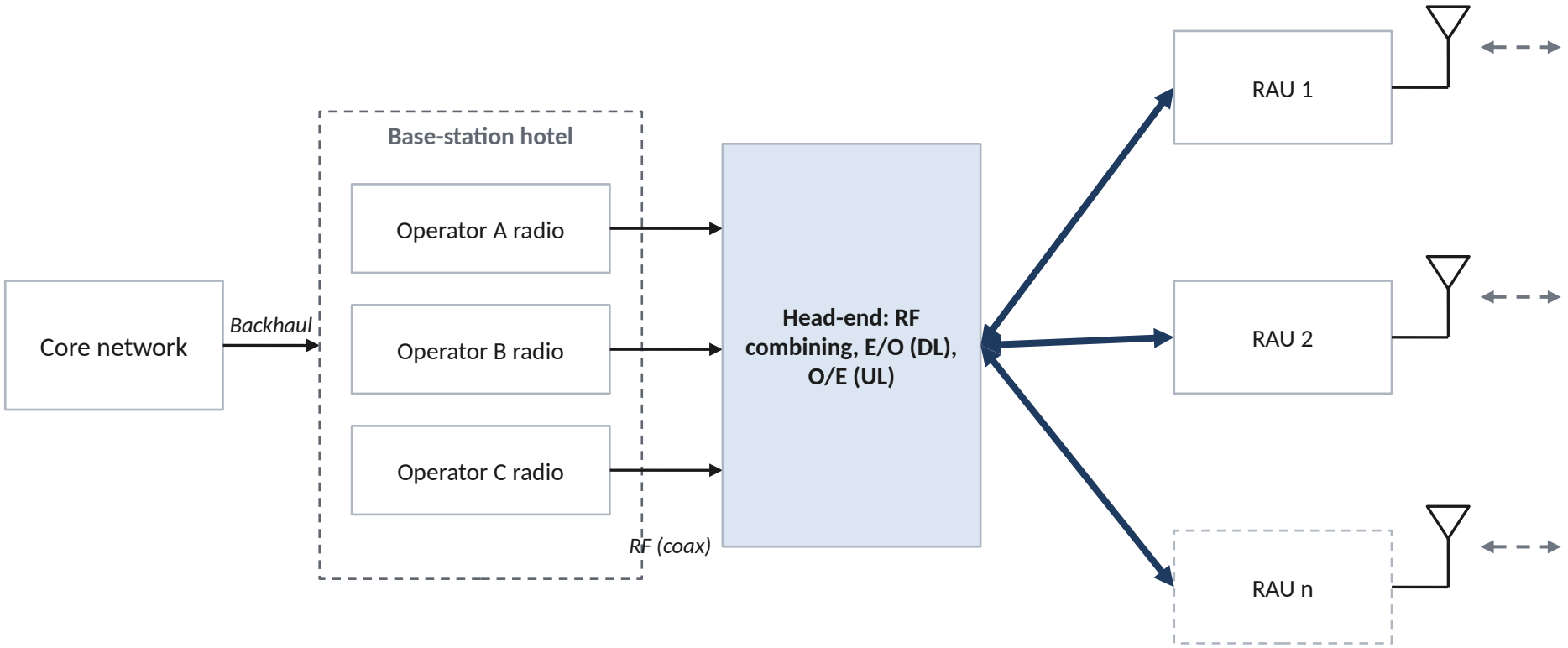
Optical loss vs RF loss
Intensity modulation with direct detection: RF power ∝ (optical power)². 1 dB optical loss → 2 dB RF loss.

Multiplexing
DL and UL use separate fibers or wavelengths. WDM is optional; needed only when several links share one fiber.

Electrical (RF / IF / digital) — Optical (fiber) — Wireless (over the air)

Sources: C. H. Cox III, Analog Optical Links: Theory and Practice, Cambridge Univ. Press, 2004; ITU-T G.Suppl.55 (10/2025). MZM: Mach-Zehnder modulator; BPF: band-pass filter; PA: power amplifier; LNA: low-noise amplifier.

Architecture: where the RoF link sits in the network



Fiber links: analog RF, downlink and uplink

Fronthaul: head-end ↔ RAU. This is the RoF link.

Backhaul: base station ↔ core; digital, packet. Not RoF.

Topology: star; one fiber pair or wavelength pair per RAU. WDM or cascaded RAUs reduce fiber count.

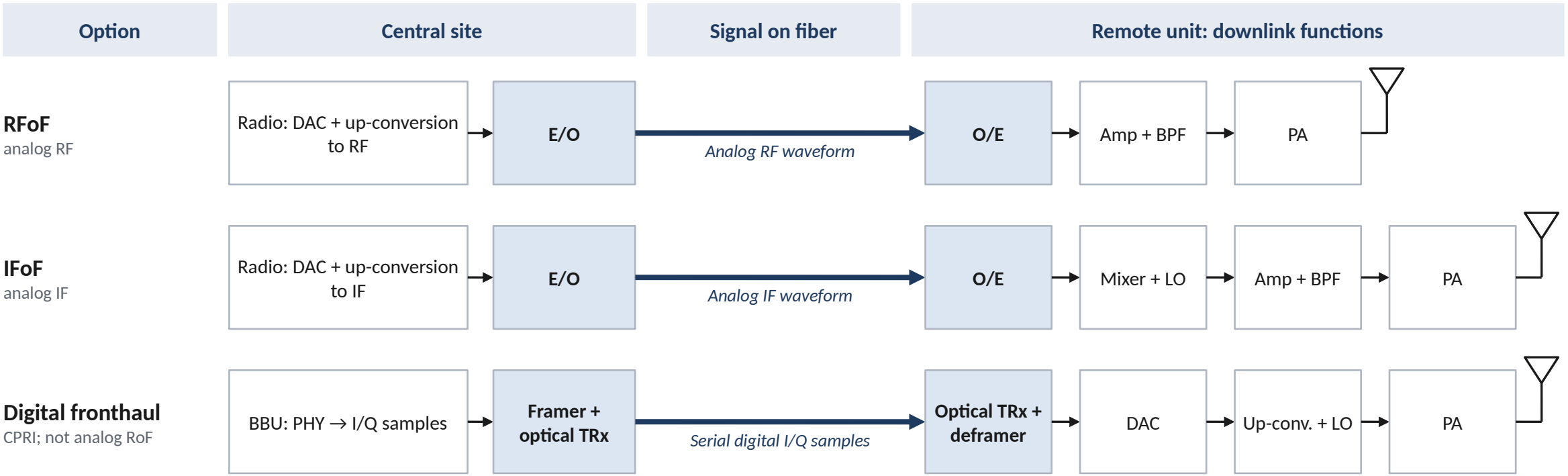
RAU contents: O/E, E/O, RF amplifiers, filters, duplexer, power supply, alarms.

C-RAN variant: BBU pool and RRHs; the link carries digital fronthaul instead.

Electrical (RF / IF / digital) — Optical (fiber) — Wireless (over the air)

Generic multi-operator fiber DAS / centralized radio architecture. Sources: ITU-T G.Suppl.55 (10/2025); ITU-T G.9803 (2018).

RoF types: what is on the fiber sets what the remote unit contains



Remote-unit complexity increases from RFoF to digital fronthaul (frequency conversion, then DAC/ADC and framing).

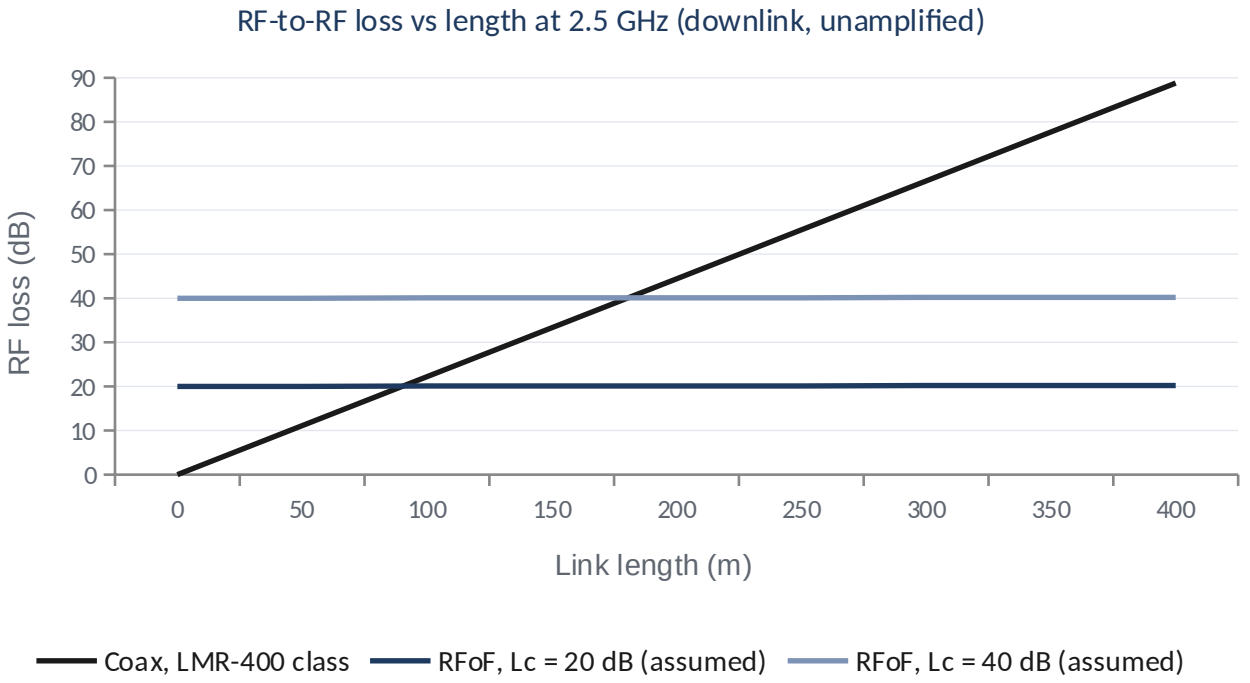
Fiber transport requirement increases in the opposite direction: digitized samples need far more capacity than the analog waveform (slide 8).

Electrical (RF / IF / digital) — Optical (fiber)

Uplink chains (LNA, E/O at remote; O/E at central) omitted for clarity.

Comparison 1: passive coaxial feeder vs analog RFoF link

Item	Definition
Systems	(1) Passive coax, LMR-400 class, 50 Ω. (2) Analog RFoF link, no RF amplifiers.
Endpoints	Head-end RF output port → remote antenna port (downlink)
Signal	RF passband at 2.5 GHz
Medium	Coax; G.652.D single-mode fiber at 1550 nm
Criterion	RF-to-RF insertion loss versus length
Assumptions	Coax 22.2 dB/100 m. Fiber 0.25 dB/km optical = 0.5 dB/km RF. RoF conversion loss Lc = 20–40 dB (assumed).



Result under these assumptions: break-even length $L^* = Lc / (0.222 - 0.0005\ dB/m) \approx 90$ m (Lc = 20 dB) to 180 m (Lc = 40 dB). Below L^*, passive coax has lower RF loss; above L^*, RFoF does. Practical fiber DAS equipment adds RF gain at both ends, so the deciding metrics become noise figure, dynamic range and power consumption, not loss alone.

Coax: Times Microwave LMR-400 datasheet, 22.2 dB/100 m at 2500 MHz. Fiber: G.652.D cable datasheets, ≤ 0.22–0.25 dB/km at 1550 nm. Lc range is an assumption: verify against the datasheet of the equipment under evaluation.

Comparison 2: analog RoF vs digital fronthaul (CPRI)

Systems: analog RFoF / IFoF vs CPRI-type digital fronthaul · Endpoints: central site ↔ remote unit · Medium: single-mode fiber · Assumption: same radio carrier and antenna count

Criterion	Analog RoF (RFoF / IFoF)	Digital fronthaul (CPRI)
Signal on fiber	Analog RF or IF waveform	Serialized time-domain I/Q samples (CPRI); packets (eCPRI)
Transport requirement	Analog bandwidth ≈ radio channel bandwidth per antenna stream, plus required dynamic range	Bit rate = sample rate × 2 × bits per sample × overheads. 20 MHz LTE, 1 antenna: 1228.8 Mb/s (CPRI option 2)
Scaling	One RF / IF channel (subcarrier or wavelength) per antenna stream	Linear in antenna count and sample rate. 100 MHz, 64 antennas ≈ 252 Gb/s before line coding (computed)
Remote unit	O/E, E/O, amplifiers, filters (+ mixer and LO for IFoF)	DAC / ADC, framing, up / down-conversion, LO, amplifiers (RRH)
Dominant impairments	Noise, nonlinear distortion, dispersion-induced RF power fading; accumulate along the link	Quantization noise, fixed by bit width; independent of length while bit errors are negligible
Monitoring and interoperability	No native bit-level monitoring; largely vendor-specific. ITU-T G.9803 sets system requirements	Bit-level monitoring; CPRI / eCPRI specifications; O-RAN split 7.2x for multi-vendor
Added latency	Fiber propagation ≈ 5 µs/km; no sampling or framing	Fiber propagation ≈ 5 µs/km + framing and buffering (implementation-specific)

Sources: CPRI Specification V7.0 (2015); eCPRI Specification V2.0 (2019); ITU-T G.9803 (2018) and Amd. 2 (2022); Y. Li et al., arXiv:2111.13851 (2021). Propagation delay from group index ≈ 1.47. eCPRI split 7.2x rate scales with MIMO layers and load, not antenna count.

Benefits, and the conditions under which they hold

Benefit	Holds when	Does not imply
Low fiber attenuation (≈ 0.2–0.25 dB/km cabled SMF at 1550 nm)	Link is longer than the break-even length set by conversion loss (slide 7)	Low RF-to-RF loss for short links; conversion loss is fixed
Fiber span immune to electromagnetic pickup; galvanic isolation	Dielectric cable is used	Immunity of RAU electronics or of the air interface
Centralized radio equipment (base-station hotel, multi-operator)	RFoF or IFoF; head-end hosts all baseband and radio processing	No remote power or maintenance: RAU has lasers, amplifiers, power supply
Format transparency: multi-band, multi-operator, multi-technology on one link	Within optoelectronic bandwidth, linearity and dynamic range	Unlimited bands: all carriers share one link's dynamic range
Small, light cable	Compared with coax of similar loss over the same route	Lower total installed cost: site-specific, not established here
Low added transport latency (≈ 5 µs/km, no sampling or framing)	Analog link	Lower end-to-end RAN latency, which baseband processing dominates

Sources: G.652.D cable datasheets (attenuation); ITU-T G.Suppl.55 (10/2025); C. H. Cox III, Analog Optical Links, 2004.

Limitations and engineering trade-offs

Noise and dynamic range: laser RIN, shot noise and receiver thermal noise set a noise floor; the link noise figure limits uplink sensitivity and the number of carriers per link.

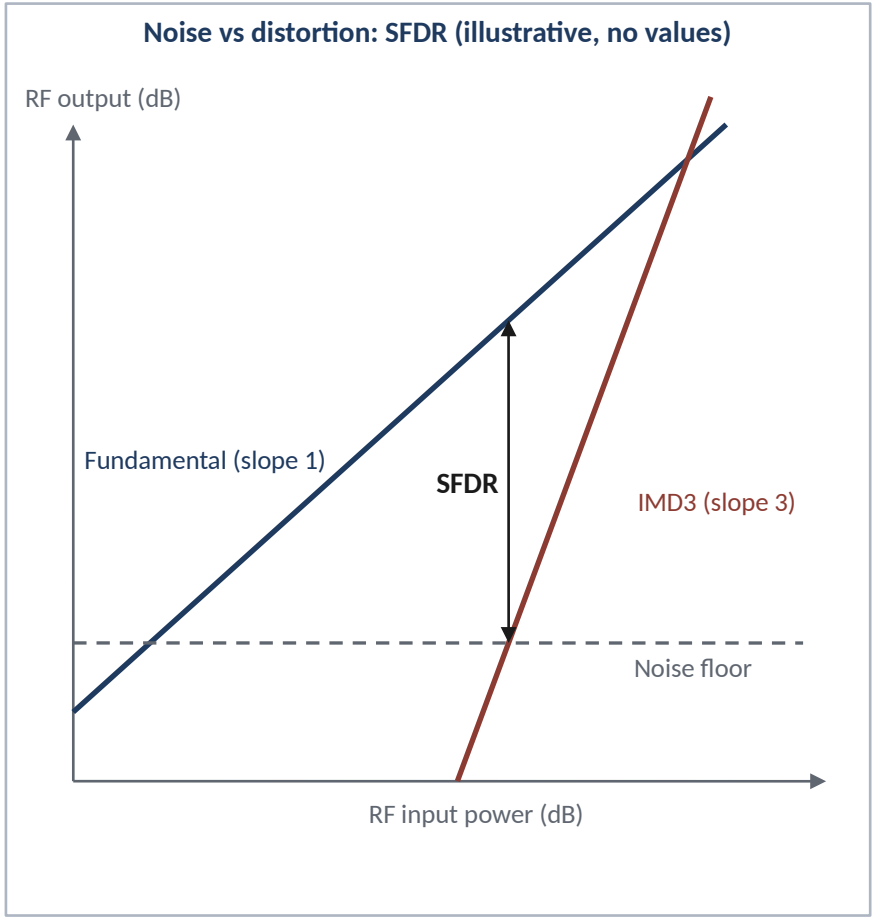
Nonlinear distortion: laser and modulator nonlinearity creates intermodulation; higher drive improves SNR until distortion dominates (SFDR).

Chromatic dispersion: with double-sideband intensity modulation, RF power fades at frequency- and length-dependent nulls. Mitigation: 1310 nm, single-sideband modulation, IFoF.

Fiber and topology: one fiber pair or wavelength pair per RAU; new fiber needs civil works and permits.

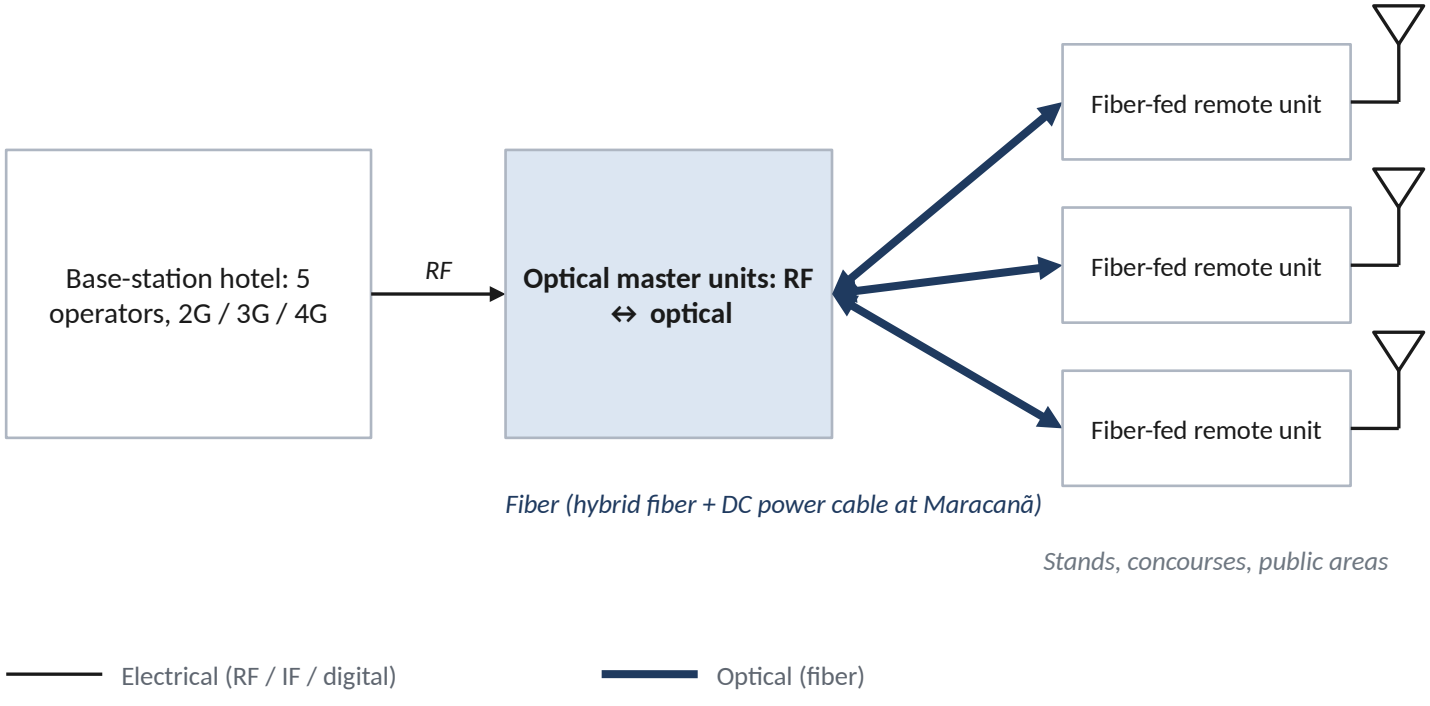
Remote power: RAU lasers and amplifiers need local power or hybrid fiber–copper cable.

Operations: no bit-level performance monitoring on the analog link; interfaces largely vendor-specific.



Sources: C. H. Cox III, Analog Optical Links, 2004; U. Gliese et al., IEEE Trans. Microwave Theory Tech. 44(10), 1996 (dispersion-induced RF power fading); D. Wake et al., J. Lightwave Technol. 28(16), 2010.

Documented case: Maracanã and Mineirão stadiums (2014)



Reported

- Optical DAS for five mobile operators, 2G / 3G / 4G; combined stadium capacity over 141,000 (Axell Wireless, June 2014).
- Maracanã: hybrid fiber + DC power cable feeding remote radio units; 2G, 3G, 4G and iDEN (RFS, 10 June 2014).

Not reported

- Link budgets, noise figure, measured capacity, cost against alternatives.
- Transport format per subsystem (analog RF or digital).

What the case supports: feasibility of a multi-operator, multi-technology fiber-fed DAS in large venues. It does not support quantitative performance or cost claims.

Sources: Axell Wireless press release, June 2014 (prnewswire.co.uk); RFS press release, 10 June 2014 (globenewswire.com); PBE Axell case study "Brazilian Sports Stadiums" (equipment names: verification required).

Conclusions

#	Conclusion	Evidence
1	RoF transports an existing radio signal. It does not add spectrum or air-interface capacity.	Slide 3
2	Against passive coax at 2.5 GHz, unamplified RFoF has lower RF loss only beyond a break-even length set by conversion loss (≈ 90–180 m under the stated assumptions).	Slide 7
3	Against digital fronthaul, analog RoF needs far less fiber capacity and no DAC/ADC at the remote unit, at the cost of noise, distortion, dispersion and weaker monitoring.	Slides 8, 10
4	Documented deployments include multi-operator fiber-fed DAS in large venues; vendor sources support feasibility, not performance.	Slide 11
5	Selection criteria: fiber availability, link length, number of bands and operators, required noise figure and dynamic range, remote power, monitoring needs.	Slides 7–10

Questions

Reference material in the appendix:

- A1 Glossary
- A2 Loss figures at 100 m, 1 km and 10 km
- A3 Remote antenna unit types (BBoF, IFoF, RFoF)
- A4 Measured BER versus carrier-to-sideband ratio
- A5 Research demonstration: 5G-NR and a candidate 6G signal on one fiber
- A6 Fronthaul bit-rate calculation and reported analog vs digital comparison
- A7 Standards and specifications
- A8 References

A1 Glossary

Term	Meaning
BBU	Baseband unit: PHY / MAC processing at the central site
RRH / RRU	Remote radio head / unit: DAC/ADC, frequency conversion, PA/LNA; fed by digital fronthaul
RAU	Remote antenna unit of an analog RoF link: O/E, E/O, RF amplifiers, filters
DAS	Distributed antenna system: one signal source, many antenna points
WDM	Wavelength-division multiplexing: several wavelengths on one fiber
MZM	Mach-Zehnder modulator: external intensity modulator
IM/DD	Intensity modulation, direct detection
LO	Local oscillator, used for frequency conversion
SMF	Single-mode fiber (ITU-T G.652 family)

Term	Meaning
TIA / PA / LNA	Transimpedance / power / low-noise amplifier
RIN	Relative intensity noise of the laser
SFDR	Spurious-free dynamic range
CSR	Optical carrier-to-sideband power ratio
BER	Bit error ratio
EVM	Error vector magnitude; 3GPP limit for 256-QAM: 3.5 %
CPRI	Common Public Radio Interface: serial digital I/Q fronthaul
eCPRI	Packet-based fronthaul; supports intra-PHY splits
C-RAN	Centralized RAN: baseband pooled at a central site

EVM limit: 3GPP TS 38.104, base-station transmitter EVM requirements.

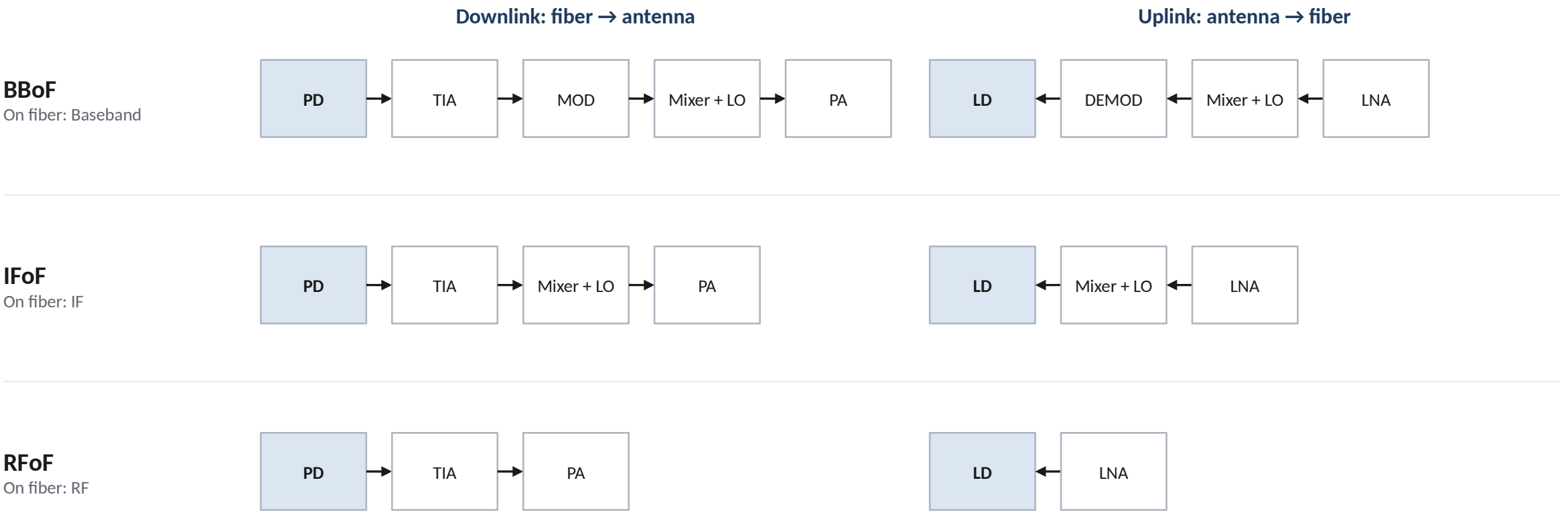
A2 Loss figures at 100 m, 1 km and 10 km (2.5 GHz)

Length	Coax, LMR-400 class (22.2 dB/100 m)	Fiber, optical loss (0.25 dB/km)	RFoF, RF loss = Lc + 2 × optical loss
100 m	22.2 dB	0.025 dB	Lc + 0.05 dB
1 km	222 dB (extrapolated; not built passively)	0.25 dB	Lc + 0.5 dB
10 km	2,220 dB (extrapolated; not built passively)	2.5 dB	Lc + 5 dB

- Lc: fixed RF-to-RF conversion loss of the unamplified RoF link; device-specific. Verification required against equipment datasheets.
- Optical-to-RF factor of 2 applies to intensity modulation with direct detection (square-law photodetection).
- Splice and connector losses are excluded from both columns.

Sources: Times Microwave LMR-400 datasheet; G.652.D cable datasheets; C. H. Cox III, Analog Optical Links, 2004.

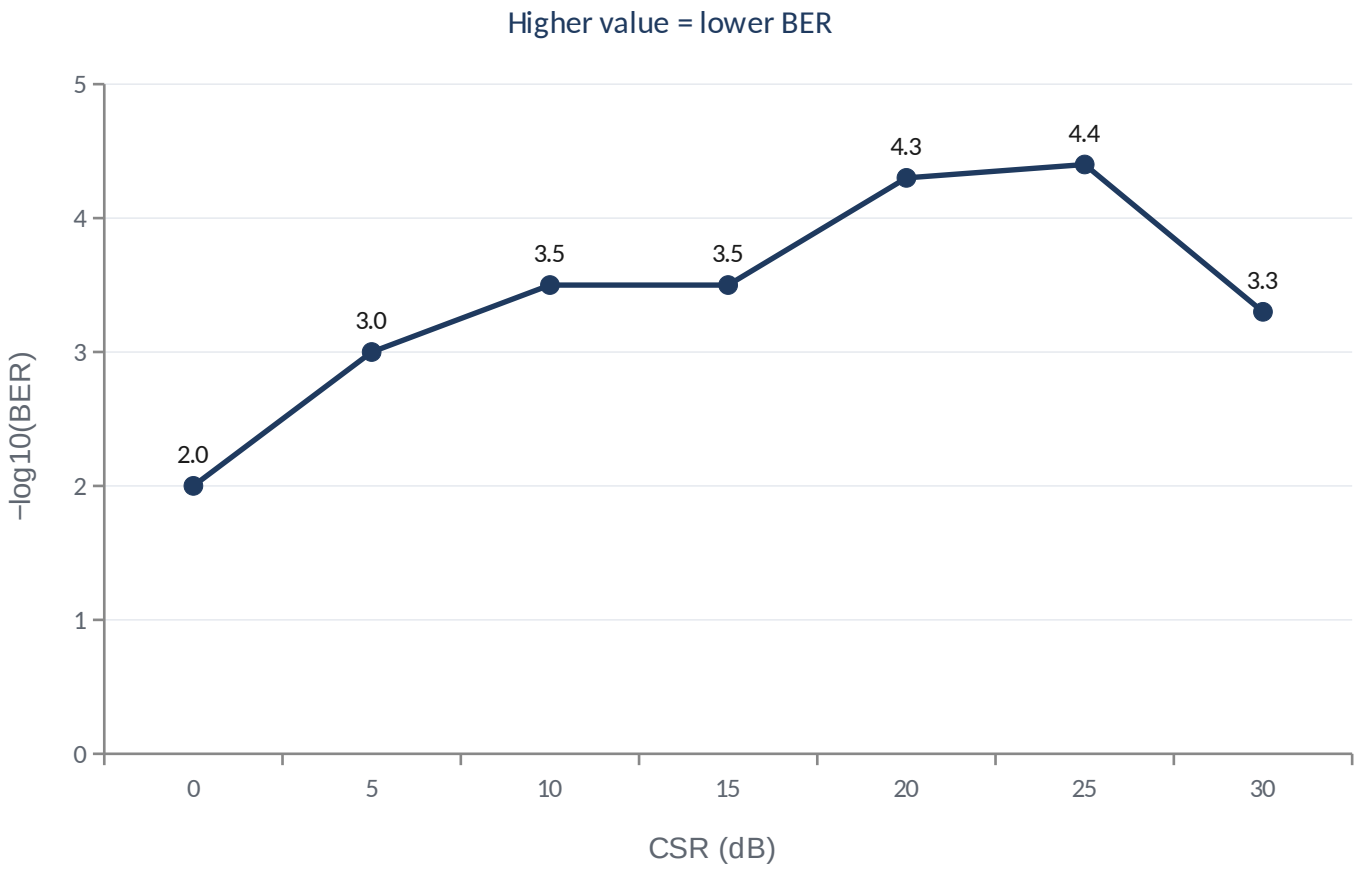
A3 Remote antenna unit types: BBoF, IFoF, RFoF



PD: photodiode · TIA: transimpedance amplifier · MOD / DEMOD: modulator / demodulator · LO: local oscillator · PA: power amplifier · LNA: low-noise amplifier · LD: laser diode. Shaded blocks are O/E (downlink) and E/O (uplink).

Redrawn from: "Radio over Fiber: An Alternative Broadband Network Technology for IoT," Electronics 9(11), 1785 (2020), Fig. 3. DOI: 10.3390/electronics9111785.

A4 Measured BER versus carrier-to-sideband ratio (one experimental link)



Setup: 10 km SMF plus 3 m wireless link (as reported).

Trend: BER improves up to CSR ≈ 25 dB, then degrades at 30 dB.

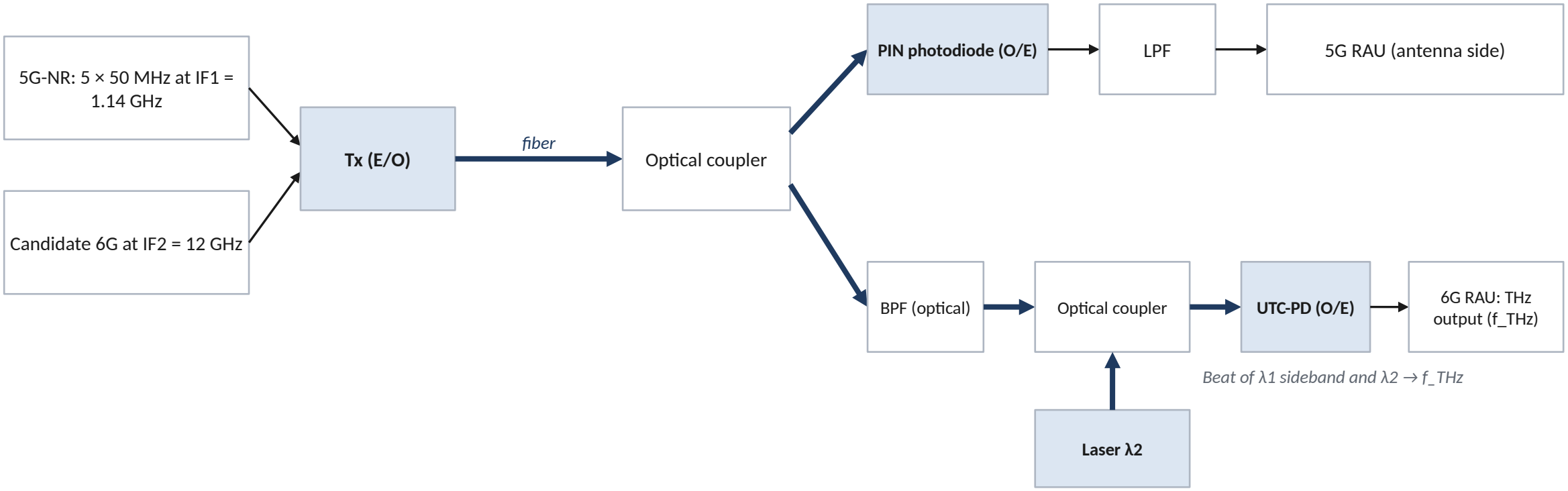
Mechanism: distortion dominates at low CSR; noise dominates at high CSR.

Limit: result applies to the tested link only.

Status: values read from the published plot; verification required.

Redrawn from: M. U. Hadi and G. Murtaza, "Fibre Wireless Distributed Antenna Systems for 5G and 6G Services," Electronics 12(1), 64 (2023), Fig. 6. DOI: 10.3390/electronics12010064.

A5 Research demonstration: 5G-NR and a candidate 6G signal on one fiber



Research demonstration of simultaneous transport. No deployment, cost or capacity conclusion follows from it.

Redrawn and simplified from: M. U. Hadi and G. Murtaza, Electronics 12(1), 64 (2023), Fig. 2. DOI: 10.3390/electronics12010064.

A6 Fronthaul bit-rate calculation and reported analog vs digital comparison

CPRI-type rate per antenna-carrier

$$R = f_s \times 2\ (I, Q) \times N \text{ bits} \times 16/15 \text{ (control words)} \times \text{line-code factor (10/8 for 8B/10B)}$$

Case	fs	N	Antennas	Result
LTE 20 MHz	30.72 Msample/s	15	1	1228.8 Mb/s with 8B/10B (CPRI option 2)
NR 100 MHz	122.88 Msample/s	15	1	3.93 Gb/s before line coding
NR 100 MHz	122.88 Msample/s	15	64	≈ 252 Gb/s before line coding

Reported modeled comparison (Li et al., 2021)

- Up to 40× lower fiber bandwidth for analog RoF than digital RoF in one modeled case: 16 × 16 MIMO, 28 GHz carrier, 256-QAM.
- In a modeled 256-QAM case, analog RoF EVM ≈ 8 % against the 3.5 % 3GPP limit; the digital case met the limit.
- Both results are scenario-specific. Values carried over from the original deck: verification against the paper required.

Sources: CPRI Specification V7.0 (2015); 3GPP TS 38.104; Y. Li, Q. Xie, M. El-Hajjar, L. Hanzo, "Analogue Radio Over Fiber for Next-Generation RAN: Challenges and Opportunities," arXiv:2111.13851 (2021).

A7 Standards and specifications

Document	Body	Scope relevant to this talk
G.9803 (2018), Amd. 1 (2019), Amd. 2 (2022)	ITU-T	Radio over fibre systems: system requirements; Amd. 2 covers RoF architectures and requirements for IMT-2020 mobile fronthaul
G.Suppl.55 (10/2025)	ITU-T	Radio-over-fibre technologies and their applications; earlier editions 07/2015, 12/2021, 12/2023
G.652	ITU-T	Characteristics of standard single-mode optical fibre and cable
CPRI Specification V7.0 (2015)	CPRI industry consortium	Serial digital I/Q interface between baseband and radio equipment
eCPRI Specification V2.0 (2019)	CPRI industry consortium	Packet-based fronthaul with intra-PHY functional splits
O-RAN fronthaul (split 7.2x)	O-RAN Alliance	Multi-vendor control, user and synchronization plane specification
TS 38.104	3GPP	NR base-station RF requirements, including transmitter EVM limits

Edition dates of G.Suppl.55 from the ITU-T recommendation page. Amendment content of G.9803 as cited in the original deck: verification against ITU-T text recommended.

A8 References

1. C. H. Cox III, Analog Optical Links: Theory and Practice, Cambridge University Press, 2004.
2. U. Gliese, S. Nørskov, T. N. Nielsen, "Chromatic dispersion in fiber-optic microwave and millimeter-wave links," IEEE Trans. Microwave Theory Tech., 44(10), 1996.
3. D. Wake, A. Nkansah, N. J. Gomes, "Radio over fiber link design for next generation wireless systems," J. Lightwave Technol., 28(16), 2010.
4. ITU-T G.9803 (2018), Radio over fibre systems; Amendment 1 (2019); Amendment 2 (2022).
5. ITU-T G-series Supplement 55 (10/2025), Radio-over-fibre (RoF) technologies and their applications.
6. ITU-T G.652, Characteristics of a single-mode optical fibre and cable.
7. CPRI Specification V7.0 (2015); eCPRI Specification V2.0 (2019).
8. 3GPP TS 38.104, NR; Base Station (BS) radio transmission and reception.
9. Y. Li, Q. Xie, M. El-Hajjar, L. Hanzo, "Analogue Radio Over Fiber for Next-Generation RAN: Challenges and Opportunities," arXiv:2111.13851, 2021.
10. "Radio over Fiber: An Alternative Broadband Network Technology for IoT," Electronics 9(11), 1785, 2020. DOI: 10.3390/electronics9111785.
11. M. U. Hadi, G. Murtaza, "Fibre Wireless Distributed Antenna Systems for 5G and 6G Services," Electronics 12(1), 64, 2023. DOI: 10.3390/electronics12010064.
12. Times Microwave Systems, LMR-400 coaxial cable datasheet.
13. Axell Wireless press release on Brazilian stadium coverage, June 2014; RFS press release on Maracanã, 10 June 2014; PBE Axell case study "Brazilian Sports Stadiums".